

ID: _____

NAME: Mamga

SECTION: _____

1. Write the expressions for 3-bits comparator for
- $A < B$
- ,
- $A = B$
- and
- $A > B$

$$x_n = A_n \odot B_n \quad [2]$$

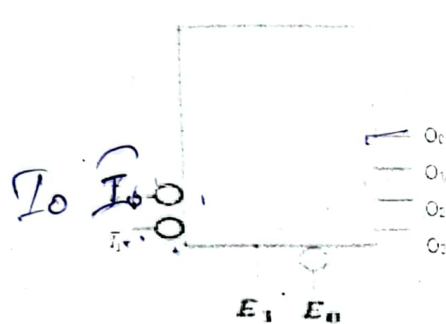
$$A < B \quad \bar{A}_2 \cdot B_2 + \bar{A}_1 \cdot B_1 \cdot x_2 + \bar{A}_0 \cdot B_0 \cdot x_2 x_1$$

$$A = B \quad A_2 \odot B_2 + A_1 \odot B_1 + A_0 \odot B_0$$

$$A > B \quad A_2 \bar{B}_2 + A_1 \bar{B}_1 x_2 + A_0 \bar{B}_0 x_2 x_1$$

2. Produce expressions for 2x4 Decoder given below? Drawing table will help

[4]



$$O_0 = E_1 \bar{E}_0 [I_0 I_1]$$

$$O_1 = E_1 \bar{E}_0 [I_0 \bar{I}_1]$$

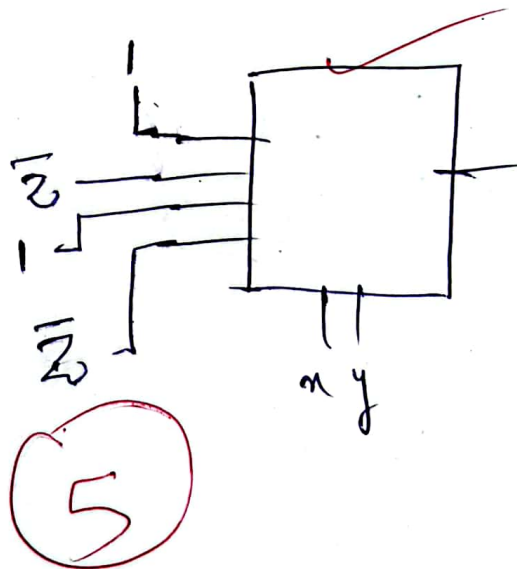
$$O_2 = E_1 \bar{E}_0 [\bar{I}_0 I_1]$$

$$O_3 = E_1 \bar{E}_0 [\bar{I}_0 \bar{I}_1]$$

3. Implement given function using 4 X 1 multiplexer?
- $F(x, y, z) = \sum(0, 1, 2, 4, 5, 6)$

Table required [5]

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



ID: 231-0810

QUIZ 3-A (EE-1005)

NAME: Hemza Tariq

SECTION: _____

4. Design Priority 4 x 2 Encoder? Create table, K-map, and expressions? $I_1 < I_3 < I_0 < I_2$

[5]

NOTE: Truth Table must follow High to Low priority (High at Left to Lowest at Right)

I_2	I_1	I_3	I_0	A	B
1	x	x	x	1	0
0	1	x	x	0	0
0	0	1	x	1	1
0	0	0	1	0	1
0	0	0	0	x	x

A

$I_2 I_1$	00	01	11	10
$I_2 I_0$	00	01	11	10
00	X	0	1	1
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

B

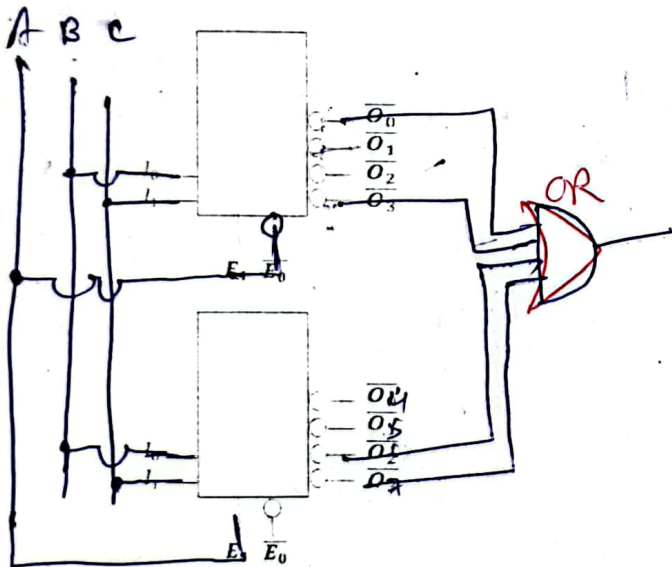
$I_2 I_1$	00	01	11	10
$I_2 I_0$	00	01	11	10
00	X	1	1	1
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$A = I_2 + \bar{I}_0 I_3$$

$$B = \bar{I}_2 \bar{I}_0$$

5. Use Decoder to Implement canonical term give below? Rename Inputs according to given function [4]

$$F(A, B, C) = \prod (0, 3, 6, 7)$$



6. Write the expressions for the 4x1 Multiplexer given below? [5]

$$F = E_1 \bar{E}_0 \left(D_0 \bar{S}_1 \bar{S}_0 + D_1 \bar{S}_1 S_0 + D_2 S_1 \bar{S}_0 + D_3 S_1 S_0 \right)$$

E_1	E_0
1	0
1	1

